Sunidhi

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EDUCATION

Duke University (CGPA: 3.7/4.0) **May 2026**
Master of Engineering Biomedical Engineering- Medical Devices Durham, NC
Courses: ADM (Advanced Manufacturing and Design), Design Health, Cardiac Electrophysiology, Quality Systems, Biomaterials, Signals

Manipal University Jaipur (CGPA: 4.0/4.0) **May 2023**
Master of Science in Biotechnology-Industrial Engineering Jaipur, IND
Courses: Bioinformatics, Biostatistics, Genetics, Material Design, Industrial Manufacturing, Biosensor design, Medical Instrumentation

EXPERIENCE

Johnson & Johnson Medtech (Electrophysiology) **September 2025- January**
NPD Quality Co-op Irvine, CA

- Designed and validated a robotic ablation actuator interfaced with the TruePulse Generator to autonomously perform 100 PFA ablation cycles per cable that eliminated manual testing, leading to a $38K cost reduction and improved precision.
- Executed Design Verification Testing (DVT) for commercial catheter builds, analyzing functional and performance data against design inputs to validate product reliability and ensure compliance with design control requirements before release.
- Authored a new standard test method (STM) for ablation performance testing, increasing measurement precision by 18% and repeatability by 22%, enabling consistent data generation and improved reliability across commercial catheter builds.
- Performed an in-depth failure investigation on catheter lots by fatigue-life testing that established thermal stress from repetitive pulsing as the root cause, suggested design improvements that reduced recurrence by 73% under worst-case conditions.
- Built end-to-end traceability for a subsystem from user needs to design outputs to verification evidence, closing 100 percent of trace gaps identified during internal review and accelerating design freeze readiness.
- Authored and maintained 18 risk management artifacts, including DFMEA and hazard analyses, closing 14 high-severity items by converting mitigations into testable design requirements with verified evidence.

Johnson & Johnson Medtech (Electrophysiology) **May 2025-August 2025**
R&D Quality intern Irvine, CA

- Developed and executed Test Method Validation (TMV) protocols for PFA ablation catheter deflection testing; performed blinded validation and Gage R&R studies on medium-risk systems, improving reproducibility by >15% and ensuring compliance with ISO 13485.
- Authored an engineering report on a catheter lot deviation labelled “use-as-is” by performing worst-case risk assessment and tolerance analyses, leading to regulatory acceptance for the lot and a specification revision that strengthened long-term quality control.
- Optimized polyurethane (PU) application process by designing controlled experiments to quantify material usage, utilized the results to update SOPs to reduce process variability by ~20% and enhance manufacturing consistency.
- Detected process drift in catheter dimensions and used Minitab control charting, normality testing, and Cp/Cpk analysis to assess variability, driving targeted process changes that improved capability by 0.2 and reduced variation by ~25%.
- Facilitated risk reviews across 3 subsystem iterations, converting failure modes into 9 design changes that lowered high-severity risk rankings and strengthened verification alignment before release.
- Co-led the CREDO Action Team and hosted a 150-plus employee campus-wide event to strengthen cross-functional collaboration alignment.

S&M Industries **May 2023-July 2024**
Manufacturing and Operations Head India

- Directed ISO controlled Class I cleanroom manufacturing across 2 production lines and 4 SKUs, managing daily documentation review, in-process quality checks, and dispatch execution to deliver ~10,000 units per month with 98% OTIF shipments and >97% FPY.
- Drove defect containment and disposition for line events, executed bracketing, initiated CAPA investigations, and implemented corrective actions with QC and Ops, reducing defect rate 35% and scrap and rework 25% in one quarter.
- Established SPC-based process controls and capability monitoring, deployed fixtures and manufacturing aids, reducing cycle time 18% and increasing equipment uptime 12%.
- Built and scaled a 14-person cross-functional floor team, implemented training certification and layered process audits, improving audit readiness and cutting escalations 30% while sustaining 98% OTIF.

PROJECT

Sleep-Apnea **September 2024- April 2025**

- Conducted over 70 hours of clinical observation and patient interviews to identify key usability and comfort challenges in existing CPAP therapy, translating unmet clinical needs into actionable design inputs for a next-generation tongue stabilization device.
- Applied structured design control and risk management (DFMEA, design reviews) to evaluate design safety, biocompatibility, and manufacturability, ensuring traceable alignment with FDA 510(k) pathways, ISO 13485 quality systems, and ISO 14971 risk standards.
- Designed, fabricated, and validated a functional prototype using biocompatible silicone and carbon-fiber composites, and testing demonstrated a 20% improvement in patient comfort and consistent airway patency under simulated clinical conditions.

Duke Stimulation Centre **October 2024- December 2025**

- Translated clinical and stakeholder needs into FDA-compliant design inputs and measurable engineering specifications for a next-generation cystoscope, ensuring full traceability through design control documentation.
- Applied Design for Manufacturability (DFM) and tolerance optimization principles to enhance injection-molded component quality and structural reliability, reducing manufacturing defects by 15% while improving consistency.

SKILLS

- **Design and Stimulation** SolidWorks, AutoCAD, Fusion360, 3D-Slicer (Medical Image Reconstruction), Arduino
- **Manufacturing** IQ, 3D-Printing, SPC, DFM, LBM, Design Control Framework, ISO13485
- **Data Analysis** Six Sigma, Regression Analysis, SQL, Pandas, NumPy, Minitab, Gage R&R, Verification & Validation
- **Mechanical &Material** Motors and sensors, Material Selection and Testing, Risk Matrix, Material Science (Polymers& Biocompatible Metals)